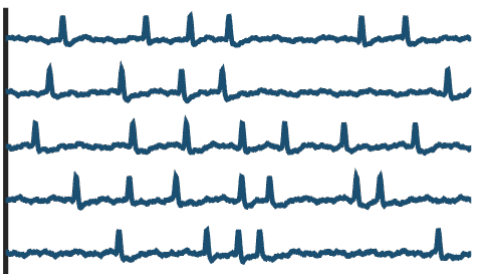


Neural Activity

Neuronal recording /
simulated network



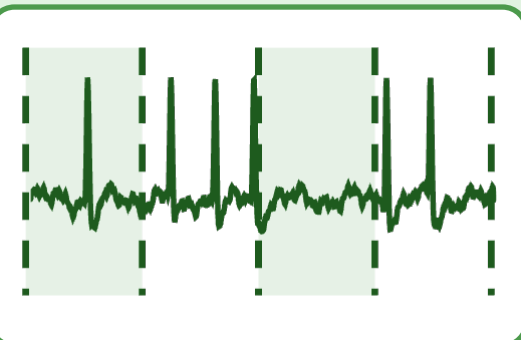
membrane potential $V_i(t)$



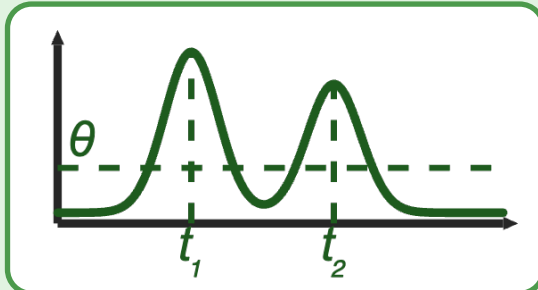
spike trains $s_i(t)$

Detect Change Points

F -test on $\text{var } \Delta^2 V_i(t)$ across
windows



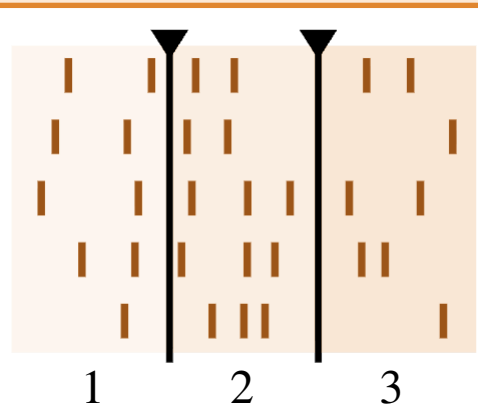
non-overlapping windows



F -statistic $\Delta^2 V_i(t)$, threshold θ

Segment Neural Activity

spike trains split at $t_1, \dots, t_{k-1}$



cut at t_1, t_2

seg 1

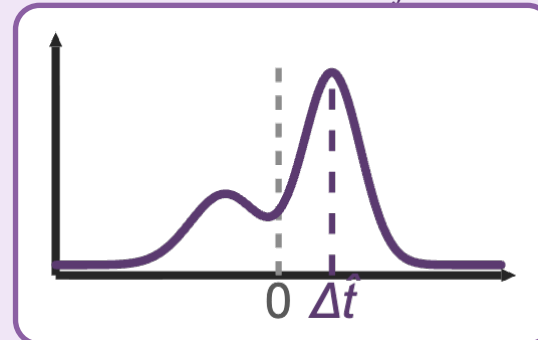
seg 2

seg 3

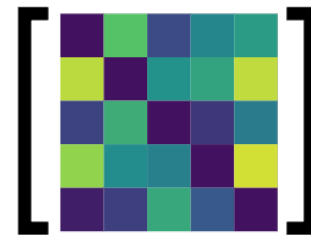
stationary segments

Infer Causal Communication

time-delayed correlation
coefficient C_{ij}



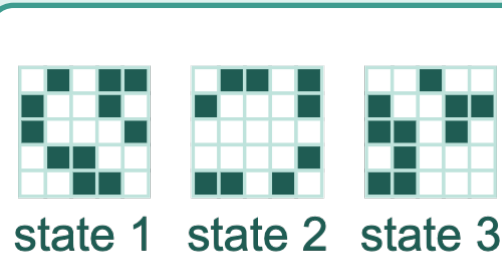
correlation vs delay



C_{ij} per segment

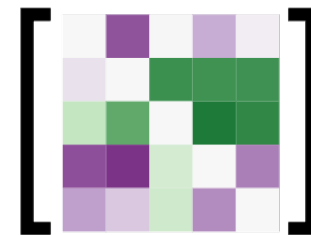
Infer Structural Connectivity

wiring change across states



state 1 state 2 state 3

W_{ij} per state



ΔW across states